

StudyGen AI

Business Requirements Specification (BRS) & Software Requirements Specification (SRS)

Version: 1.1

Tagline: Scan • Learn • Revise • Succeed

Update: Database and file-storage architecture revised from Firebase/Cloud Storage to a local-first model with MongoDB Atlas as the application database. Original user PDFs are not stored permanently on the server.

This document is the working reference for implementation of StudyGen AI Version 1.

PART A — BRS

1. Business Objective

StudyGen AI is an AI Powered Study Scanner & Smart Learning App designed as an all-in-one study tool for scanning study material, creating AI-based learning content, working with PDFs, practicing with quizzes and flash cards, and solving doubts through AI Chat.

2. Problem Statement

- Books se notes banana time-consuming.
- Revision difficult.
- PDF padhna boring.
- Important questions identify nahi hote.
- Different apps use karni padti hain.

3. Solution

- Documents Scan karna.
- AI se Notes banana.
- PDF Summary dena.
- Quiz banana.
- Flash Cards banana.
- AI Chat se doubts solve karna.

4. Main Business Features

Smart Document Scanner: Camera Scan, Gallery Import, Auto Edge Detection, Auto Crop, Manual Crop, Perspective Correction, HD Scan, Multi Page Scan, Flashlight, Offline Scan.

PDF Creator: Create PDF, Multi Page PDF, Rearrange Pages, Delete Pages, Paper Size, Organize Folder, Search Documents, Export PDF, Export JPG. Created PDFs are stored locally on the user's device by default.

AI Study Assistant: Short Notes, Detailed Notes, Summary, Key Points, Important Questions, Formula Sheet, Revision Notes.

AI Learning Features: Quiz Generator, Flash Cards, AI Chat, Explain in Easy Language, Hindi / English, Read Aloud, Ask Questions, Homework Help.

PDF AI: User selects a local PDF → temporary upload for AI/OCR processing → Summary → Quiz → Notes → Flash Cards → results saved locally and/or as user-selected app data; temporary uploaded PDF is deleted after processing.

5. History, Profile & Settings

- History: Save, Recent Notes, Recent PDF, Recent Quiz, Recent Scan, Search, Delete, Filter. History metadata and saved AI content may be stored in MongoDB; original PDFs remain local unless a future cloud-backup feature is enabled.
- Profile: Photo, Name, Email, Premium, Edit, Logout, Subscription, Achievements.
- Settings: Dark Mode, Language, Notifications, Privacy Policy, Contact, About App, Feedback, Rate App.

6. Premium & Revenue

- Premium: Unlimited AI, Unlimited Scan, Unlimited PDF, Cloud Backup, No Ads, Priority AI, Advanced Filters.
- Revenue model: Free Version → Ads → Premium Subscription → AI Credits → PDF Tools → Cloud Storage.

Cloud Backup is treated as a future/optional feature. Version 1 uses local-first storage for original PDFs.

7. User Flow

Splash → Login → Home → Scan/Create PDF → Optional AI Processing → Notes/Quiz/Revision → History

AI PDF flow: Local PDF → Temporary Upload → OCR / AI Processing → AI Chat / Study Outputs → Save results → Delete temporary server copy.

8. Future Features

- Voice Notes
- Live AI Tutor
- Handwriting Recognition
- Math Solver
- Diagram Explanation
- Mind Maps
- Study Planner
- Attendance Tracker

PART B — SRS

1. System Overview

StudyGen AI will use Flutter for the mobile application, MongoDB Atlas for persistent application data, local device storage for original user PDFs and generated files, temporary server-side file handling for AI/OCR processing, an AI provider such as Gemini/Grok, OCR, Camera Scanner and PDF Generator.

2. Technology Stack

Component	Version 1 Decision
Frontend	Flutter
Persistent Database	MongoDB Atlas Free tier during development
Original PDF / generated file storage	Local device storage by default
Temporary AI file handling	Temporary server-side storage; delete after processing
AI	Gemini / Grok API — final provider to be selected
OCR	OCR technology/package — to be finalized
Scanner	Camera Scanner
PDF	PDF Generator
UI	Material Design 3
Authentication	Authentication approach to be finalized; user identity must be linked to MongoDB records

3. Screen Requirements

- 1. Splash
- 2. Login
- 3. Home
- 4. Smart Document Scanner
- 5. Scan Preview & Crop
- 6. AI Study Assistant
- 7. AI Learning Features
- 8. PDF AI
- 9. History
- 10. Profile
- 11. Settings
- 12. Premium

4. Functional Requirements

FR-01 — Authentication: The application shall provide a Login stage and associate each authenticated user with a unique MongoDB user record. The final authentication mechanism shall be finalized before implementation.

FR-02 — Smart Document Scanner: The scanner shall support camera scan, gallery import, edge detection, crop, perspective correction, HD scan, multi-page scan, flashlight and offline scan.

FR-03 — Local PDF Creator: The application shall support PDF creation, multi-page handling, page rearrangement/deletion, paper-size selection, organization/search and PDF/JPG export. Original created PDFs shall be stored locally on the user's device by default.

FR-04 — AI Study Assistant: The application shall generate short notes, detailed notes, summary, key points, important questions, formula sheet and revision notes.

FR-05 — AI Learning Features: The application shall provide quiz generation, flash cards, AI Chat, easy-language explanation, Hindi/English, read aloud, ask questions and homework help.

FR-06 — PDF AI Temporary Processing: When a user chooses to use AI with a local PDF, the application shall upload the PDF only for the required processing session. The system shall perform OCR/AI processing and delete the temporary uploaded PDF after processing, subject to error/retry handling.

FR-07 — AI Chat & Revision: AI chat messages and user-saved AI outputs such as notes, quizzes and flashcards shall remain available for later revision according to the user's history settings. Deleting the temporary source PDF shall not automatically delete the saved AI conversation or study outputs.

FR-08 — History: The application shall provide recent notes, PDFs, quizzes and scans with search, delete and filter functions. For locally stored PDFs, History shall maintain local references/metadata rather than requiring permanent server copies.

FR-09 — Profile: The application shall provide profile information and planned edit, logout, subscription and achievements areas.

FR-10 — Settings: The application shall provide dark mode, language, notifications, privacy policy, contact, about, feedback and rate app.

FR-11 — Premium: The application shall provide the planned premium feature area and subscription concept, including future cloud backup capability.

5. Non-Functional Requirements

- Usability: Modern, student-friendly Material Design 3 interface.
- Consistency: Common design language across all screens.
- Performance: Scanning, OCR, AI and PDF operations should provide appropriate processing states.
- Security: Authentication and user data must be protected; temporary uploaded files must not be retained longer than required.
- Privacy: Original PDFs are local by default and are uploaded only when the user invokes an AI feature requiring server-side processing.
- Scalability: Architecture should allow future cloud backup and additional features.
- Offline: Scanner and local PDF creation are planned to support offline operation where technically possible.

6. Data & Backend

MongoDB Atlas shall be the persistent application database for Version 1 development. The exact collections, fields and indexes must be finalized before database implementation.

Collection	Purpose
users	User profile, account metadata, premium/subscription status and preferences.
ai_chats	AI conversation sessions, messages, titles and timestamps needed for revision.
notes	User-saved notes, summaries, key points and revision material.
quizzes	Generated/saved quizzes and questions.
flashcards	Generated/saved flash cards.

Collection	Purpose
history	References/metadata for recent user activities; local file references where applicable.
usage / credits (if required)	AI usage, credits and plan limits.

File-storage rule: MongoDB is not used as the permanent storage location for original PDFs or large binary scan files. Such files remain on-device by default. Temporary server-side copies used for AI/OCR are deleted after the required processing lifecycle.

7. AI Processing Flow

Local Document/PDF → Temporary Upload → OCR / AI Read → AI Processing → Notes / Summary / Key Points / Questions / Formula Sheet / Revision Notes / Chat → Save selected results → Delete temporary uploaded PDF.

8. Development Plan

- Phase 1: UI (Completed in Stitch)
- Phase 2: Flutter UI
- Phase 3: Authentication + MongoDB data layer
- Phase 4: Document Scanner + OCR
- Phase 5: AI Integration
- Phase 6: Local PDF Generation + temporary AI file processing
- Phase 7: Testing
- Phase 8: Play Store Release

9. Pre-Implementation Decisions

- Confirm MongoDB Atlas Free tier for development and define upgrade criteria.
- Finalize authentication mechanism and user-to-MongoDB identity mapping.
- Finalize AI provider: Gemini or Grok.
- Finalize OCR technology/package.
- Define exact MongoDB collections, fields, indexes and retention rules.
- Define temporary upload deletion/error-retry policy.
- Finalize premium pricing/subscription rules.
- Decide which future features belong to Version 2.
- Define optional Cloud Backup architecture separately from the local-first Version 1 architecture.

10. Version 1 Acceptance

- Core screens implemented.
- Navigation follows the planned flow.
- Scanner and local PDF functions connected.
- OCR and AI integrated.
- AI study outputs generated.
- AI Chat/history supports later revision.
- Temporary uploaded PDFs are deleted after successful processing.

- History/Profile/Settings/Premium connected.
- Testing completed before Play Store release.

End of StudyGen AI BRS & SRS — Version 1.1